E-COMMERCE PLATFORM

Complete Project Planning | Admin Panel + Storefront | Single-Vendor | Version 1.0 - October 2026

Project scope: A new e-commerce platform planned from scratch. The previously discussed e-commerce template plan is excluded. Multi-vendor marketplace and B2B/wholesale modules are excluded.

1. System Architecture

Two independently deployable Next.js applications (Admin Panel and Storefront) share a NestJS backend API, PostgreSQL database, Redis cache/queue, and object storage. Business rules live in the backend, not duplicated in the user interfaces.

Layer	Responsibility
Admin Panel	Products, orders, inventory, marketing, CMS, finance, staff and settings
Storefront	Shopping, product discovery, checkout, account, tracking and returns
Backend API	Authentication, authorization, validation, pricing, order state, payments, integrations
PostgreSQL	Transactional business records and reporting data
Redis + workers	Caching, scheduled tasks, webhooks, notifications and retries
Object storage	Product images, documents and uploaded media

2. Technology Stack

Area	Chosen technology
Frontend applications	Next.js, React, TypeScript, Tailwind CSS, shadcn/ui
Backend	NestJS, TypeScript, REST API
Database / ORM	PostgreSQL + Prisma
Client data / forms	TanStack Query, Zustand, React Hook Form, Zod
Jobs and cache	Redis + BullMQ
Files / media	S3-compatible object storage
Charts / analytics UI	Recharts
Monorepo	pnpm + Turborepo
Testing / delivery	Vitest, Playwright, Docker, CI/CD

3. Monorepo Structure

apps/admin, apps/storefront, apps/api, apps/worker; packages/database, packages/ui, packages/types, packages/validation, packages/api-client, packages/config; infrastructure/docker, infrastructure/monitoring; docs/architecture, docs/api, docs/database, docs/features; tests; .github.

4. Admin Panel: 22 Modules

01 Dashboard - Revenue, orders, profit estimates, KPIs, activity, charts, alerts, configurable widgets.
02 Products - Simple/variable/digital/bundle products, SKU, media, pricing, bulk edits, import/export.
03 Catalog - Categories, brands, attributes, category ordering, filters and metadata.
04 Inventory - Warehouses, stock movements, reservations, transfers, adjustments, low-stock alerts.
05 Orders - Order list, details, status timeline, fulfillment, invoices, split/partial actions.
06 Customers - Profiles, addresses, segments, tags, history, consent and customer support notes.
07 Payments - COD, supported gateways, transactions, refunds, reconciliation and webhook logs.
08 Shipping - Courier adapters, zones, rates, labels, tracking, COD and return-to-origin.
09 Returns - Return requests, inspections, refunds, replacements, exchanges and warranties.
10 Promotions - Coupons, automatic discounts, flash sales, BOGO, rules and limits.
11 Marketing - Campaigns, SMS/email/WhatsApp integrations, abandoned carts and attribution.
12 CMS - Pages, homepage sections, banners, menus, blog, content preview and scheduling.
13 Themes - Branding, colors, fonts, layouts, preview, versioning and rollback.
14 SEO & Search - Metadata, sitemap, redirects, schema, synonyms, ranking and search analytics.
15 Reviews - Verified reviews, ratings, moderation, replies, photos and product Q&A.
16 Finance - COGS, expenses, fees, settlements, tax settings, profit and loss.
17 Reports - Sales, product, customer, inventory, shipping, payments and custom exports.
18 Staff & Roles - Staff users, role-based permissions, approval rules, sessions and audit trail.
19 Notifications - Templates, alerts, channel preferences, retries and delivery logs.
20 Purchasing - Suppliers, purchase orders, goods receipts, invoices and cost history.
21 Settings - Store profile, locale, payments, shipping, integrations, API keys and webhooks.
22 System - Security, background jobs, backup status, monitoring and operational logs.

5. Storefront Pages and Capabilities

Page / Area	Key functions
Homepage	Hero, featured categories/products, banners, offers, CMS sections
Shop / Category / Brand	Product listing, filters, sorting, pagination and search
Product detail	Gallery, variants, price, stock, reviews, related items, add to cart
Search	Suggestions, typo handling, search results and filters
Cart / Checkout	Guest cart, coupons, shipping calculation, addresses, payment and COD
Order confirmation / tracking	Order number, timeline, shipment tracking and notifications
Customer account	Registration/login, profile, addresses, order history and wishlist
Returns / reviews	Return request, refund status, product feedback
Content pages	About, contact, FAQ, blog, privacy, terms and policies

Storefront UX: Responsive and mobile-first, quick add-to-cart, buy-now, persistent cart, optimized images, accessible forms, skeleton loading, SEO-friendly URLs and Bangladesh district/area selection.

6. Database Domain Groups

Domain	Representative entities
Catalog	products, product_variants, categories, brands, attributes, product_media
Orders	orders, order_items, order_events, fulfillments
Customers	customers, customer_addresses, wishlists, sessions
Payments	payments, payment_transactions, refunds, settlements
Inventory	warehouses, stock_levels, stock_reservations, stock_movements
Shipping	shipments, courier_integrations, shipping_zones, shipping_rates
Marketing	coupons, promotions, campaigns, carts
CMS	pages, page_sections, banners, menus, themes
Operations	staff_users, roles, permissions, audit_logs, notifications

Before implementation, define each table's columns, relationships, constraints, indexes, lifecycle, deletion policy and migration plan.

7. Critical Business Workflows

• Product publishing: Admin creates/edits product -> backend validates -> product and variants saved -> published items appear on storefront.

• Order creation: Customer checks out -> backend recalculates prices/shipping/discounts -> creates order and inventory reservation -> verifies payment or records COD -> confirms order.

• Fulfillment: Admin processes -> picks/packs -> courier consignment created -> tracking events synced -> delivered/failed/RTO recorded.

• Returns: Customer requests return -> eligibility check -> approval and inspection -> stock decision -> refund/exchange -> finance reconciliation.

• Reliability: Payment and courier webhooks must be authenticated, idempotent and retryable. Order and inventory transitions must be audited.

8. Bangladesh-Specific Integrations

Design adapters for bKash, Nagad, SSLCommerz, aamarPay and COD where merchant/API access is available. Courier adapters may support Pathao Courier, Steadfast, RedX, eCourier and Paperfly, subject to each provider's API access and commercial terms. Support BDT, local phone/address formats, delivery zones and COD reconciliation.

9. Security and Quality Requirements

• Server-side authorization and role-based access control; protected admin routes; secure sessions and MFA for staff.

• Input validation, rate limiting, CSRF/XSS defenses, secure uploads and secret management.

• Database transactions for orders, payments and stock; audit logs and idempotency keys.

• Automated tests for checkout, discounts, stock, refunds, permissions and webhooks.

• Backups, restore drills, error monitoring, logs, metrics and health checks.

10. Development Roadmap

Phase	Deliverables
1. Foundation	Monorepo, app scaffolds, API, environment, Docker, CI, shared packages
2. Data and auth	Schema, migrations, admin/customer authentication, RBAC
3. Admin foundation	Sidebar, layouts, dashboard, reusable tables and forms
4. Catalog and stock	Products, variants, categories, inventory and warehouse workflows
5. Storefront	Homepage, listing, detail, search, responsive layouts
6. Commerce flow	Cart, checkout, order creation, reservations, fulfillment
7. Integrations	Payments, COD, courier tracking, notifications and reconciliation
8. Full modules	CMS, promotions, marketing, finance, reports, returns, settings
9. Release	Performance, security and E2E testing, backups, deployment and monitoring

11. Launch Acceptance Checklist

• Product and stock changes sync correctly between admin and storefront.

• Checkout totals are recalculated securely on the server.

• Payment callbacks cannot create duplicate payments or orders.

• Failed/cancelled orders release stock according to defined rules.

• Courier and COD settlements can be reconciled.

• Admin permissions are enforced by backend APIs.

• Core customer journeys work on mobile and desktop.

• Backups, monitoring and rollback procedures are verified.

Scope note: Single-vendor only. No multi-vendor marketplace, vendor payouts/commissions, B2B quotations or wholesale account workflows. Multi-warehouse and multiple staff roles remain in scope.